

quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets

Qi-Xuan Yue ^a, Yufei Shen^a, Asier Larrea-Sebal ^{c,d}, Chengxin Dai ^b, Henry Webel ^e, Jose Nimo ^{g,l,m}, Fabian Coscia ^{g,m}, Orhun Kok ^h, Nikolai Slavov ^{h,i}, Juan Antonio Vizcaíno ^c, Timo Sachsenberg ^{j,k}, Mingze Bai ^a, Vadim Demichev ^f, Yasset Perez-Riverol ^{c,*}

^aChongqing Key Laboratory of Big Data for Bio Intelligence, Chongqing University of Posts and Telecommunications, Chongqing, China

^bState Key Laboratory of Proteomics, Beijing Proteome Research Center, National Center for Protein Sciences (Beijing), Beijing Institute of Life Omics, Beijing, China

^cEuropean Molecular Biology Laboratory, European Bioinformatics Institute (EMBL-EBI), Wellcome Genome Campus, Hinxton, Cambridge, UK

^dDepartment of Biochemistry and Molecular Biology, Universidad del País Vasco UPV/EHU, Bilbao, 48080, Spain

^eNovo Nordisk Foundation Center for Protein Research, Faculty of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark

^fQuantitative Proteomics laboratory, Charité – Universitätsmedizin Berlin, Berlin, Germany

^gSpatial Proteomics Group, Max-Delbrück-Center for Molecular Medicine in the Helmholtz Association (MDC), Berlin, Germany

^hDepartment of Bioengineering, Department of Biology, Department of Chemistry and Chemical Biology, Single Cell Proteomics Center, and Barnett Institute for Chemical and Biological Analysis, Northeastern University, Boston, Massachusetts, USA

ⁱParallel Squared Technology Institute, Watertown, Massachusetts, USA

^jDepartment of Computer Science, Applied Bioinformatics, University of Tübingen, Tübingen, Germany

^kInstitute for Bioinformatics and Medical Informatics, University of Tübingen, Tübingen, Germany

^lCharité – Universitätsmedizin Berlin, corporate member of Freie Universität Berlin and Humboldt-Universität zu Berlin, Berlin, Germany

^mGerman Cancer Consortium (DKTK), Partner Site Berlin, Heidelberg, Germany

Abstract

Public proteomics archives now hold thousands of data-independent acquisition (DIA) datasets, but reusing them is difficult: each was processed with a different software configuration, and most lack standardised metadata. We present **quantmsdiann**, an open-source Nextflow/nf-core workflow that runs DIA-NN in parallel across cloud and high-performance computing (HPC) infrastructure, guided by the experimental design declared in SDRF format. The workflow ships container-pinned profiles for each supported DIA-NN version through Docker and Singularity, needs no manual installation or dependency resolution, and reads all major vendor formats and the HUPO-PSI standard mzML natively. It exports harmonised quantification tables as MSstats input, the Quantitative Proteomics eXchange format (QPX), and a **pmultiqc** quality-control report. The parallel design reanalyses a 2,300-run single-cell dataset in 2.2 hours on 300 HPC nodes. Across four ProteoBench DIA instrument modalities, distributed **quantmsdiann** reproduces single-machine community submissions independently of DIA-NN release. Reanalysing public DIA deposits with a current DIA-NN build matches or exceeds the originally deposited counts, identifying 17% to 59% more protein groups per cohort, and more precursors wherever a DIA-NN-based original count is

*Corresponding author.

URL: yasset@ebi.ac.uk (Yasset Perez-Riverol )

available. `quantmsdiann` is a step towards scalable, reproducible reanalysis of the growing DIA data archive and a foundation for large-scale DIA meta-analysis and atlas building. It is available at <https://github.com/bigbio/quantmsdiann>.

Keywords: DIA-NN, single-cell proteomics, spatial proteomics, phosphoproteomics, plexDIA, data-independent acquisition, Nextflow, nf-core, SDRF, reproducibility, QPX, pmultique

1. Introduction

Data-independent acquisition (DIA) is a rapidly growing mass-spectrometry strategy for proteomics at scale, now approaching parity with data-dependent acquisition among public proteomics deposits, and DIA-NN [1] is among the most widely adopted analysis engines across timsTOF [2, 3], Astral [4], and Orbitrap platforms. Recent benchmarks confirm DIA-NN’s competitiveness for single-cell DIA workflows [5], and recent releases have expanded support for plexDIA [6, 7], timsTOF parallel accumulation-serial fragmentation (PASEF), and vendor-native raw-file reading. At the same time, DIA dataset sizes and run counts are growing rapidly [8], and the proteomics community is increasingly interested in reanalysing public deposits to recover additional identifications, integrate across cohorts, and build large-scale DIA atlases [9, 10]. DIA-NN’s command-line interface, simplicity, and analytical power make it a natural engine for mining public archives, but realising it at scale requires distributing the computation: extracting more information from public data means reprocessing cohorts of hundreds to thousands of runs fast enough that reanalysis can be iterative rather than a one-off effort. The same need extends beyond reanalysis: laboratories generating their own large cohorts increasingly have access to cloud and high-performance computing (HPC) infrastructure and benefit from distributing the analysis rather than running it on a single workstation. The steps of a DIA-NN run are heterogeneous and map naturally onto heterogeneous hardware: many lightweight per-file passes that fit small nodes, and a few memory- and compute-heavy cohort-level steps that need large nodes. Efficient large-scale analysis therefore depends on software containers and workflow engines that can place each task on appropriate cloud or HPC resources [11]. Despite this analytical maturity, most DIA-NN users run the tool as a desktop application, which limits its scalability for en masse processing of very large datasets.

In 2024 we published `bigbio/quantms` [9], a Nextflow pipeline that standardised multi-engine DIA and DDA analyses behind a Sample and Data Relationship Format (SDRF) [12] input on the `nf-core` framework [13]. `quantms` was, however, built around DIA-NN 1.8.1 with a fixed parameter set, and its DDA/DIA combination within the same workflow made extending DIA-specific functionality difficult. New DIA-NN parameters conflicted with DDA options, versions were hard-coded, sample-level acquisition parameters were not properly propagated to per-run DIA-NN invocations, and raw-file conversion was mandatory. Iterative reanalysis at archive scale requires a workflow that combines the reproducibility of `nf-core`, the sensitivity of current DIA-NN releases, the metadata discipline of SDRF, and a parallelisation strategy capable of processing thousands of raw files.

Here, we present `quantmsdiann`, an independent SDRF-driven Nextflow pipeline dedicated to DIA-NN that extends DIA-specific features beyond what the original `bigbio/quantms` pipeline could accommodate. While `quantmsdiann` shares its SDRF-first design philosophy and parts of the configuration-generation tooling with `quantms`, its workflow, container set, and module library are developed and released independently, with container-pinned profiles for every DIA-NN version selectable at runtime under Docker or Singularity. Downstream, it exports a harmonised QPX (Quantitative Proteomics eXchange) Parquet archive alongside the standard DIA-NN reports and MSstats input. We demonstrate the pipeline along three axes: ProteoBench [14] benchmarks across

four DIA modalities and successive DIA-NN releases; independent reanalyses of public DIA deposits spanning bulk cell lines (NCI-60, ProCan), single-cell datasets, a spatial Deep Visual Proteomics cohort, and a phosphoproteomics cohort; and a pan-cohort of five public DIA cell-line reanalyses integrated under a single SDRF-driven invocation. Across the single-cell and bulk cell-line cohorts, reanalysis with the current DIA-NN build identifies on average $\sim 36\%$ more protein groups ($+17\%$ to $+59\%$ across cohorts) than the original or 1.8.1 analyses, so reprocessing archived data with an up-to-date engine yields measurably more from each deposit.

2. Experimental Procedures

2.1. Pipeline implementation

`quantmsdiann` is implemented in Nextflow DSL2 ($\geq 25.10.4$) and follows nf-core community standards [13] for pipeline structure, testing, documentation, and continuous integration. It supports both DIA-NN library-free analysis (predicting the spectral library in-silico from a FASTA reference) and library-based analysis against a user-supplied spectral library; all analyses in this study used the library-free mode; the reanalysis cohorts used the UniProt human reference proteome [15] (UP000005640, downloaded March 2026), while the ProteoBench HYE benchmarks used the corresponding three-species (human, yeast, *E. coli*) FASTA. Each computational step is packaged as a versioned container image (Docker, Singularity, or Apptainer). Open-source community tools (ThermoRawFileParser [16], pridepy [17], QPX, and the `wiffconverter` SCIEX reader) are pulled from public registries (BioContainers [18] or `ghcr.io/bigbio`). Because the DIA-NN license restricts redistribution, only version 1.8.1 is shipped as an image (the default, from BioContainers); releases from 1.9 onward (1.9.2–2.5.1, including the 2.5.1-enterprise build) are built locally from the per-release recipes in the companion `quantms-containers` repository (<https://github.com/bigbio/quantms-containers>) and tagged so the workflow selects them automatically (Supplementary Note 1). The repository at <https://github.com/bigbio/quantmsdiann> is organised as a standard nf-core pipeline with main workflow (`main.nf`), subworkflows, modules, test data and CI configurations, and documentation. Pipeline version v2.2.0 (release codename “Chongqing”, 2026-06-16) was used for the analyses reported in this paper.

2.2. SDRF extensions for `quantmsdiann`

`quantmsdiann` accepts SDRF [12] as primary input. SDRF parsing is performed using the `sdrf-pipelines` Python package developed within the `quantms` project, which validates the SDRF against the proteomics-specific subset of EFO, PSI-MS, and PRIDE ontology terms and emits a normalised per-sample channel for downstream Nextflow processes. To express DIA- and DIA-NN-specific acquisition settings at the sample level, we extended the SDRF-Proteomics format (version 1.1.0 [12]; specification and templates at <https://github.com/bigbio/proteomics-sample-metadata>) with additional comment columns: the precursor (MS1) and fragment (MS2) m/z scan ranges and, for multiplexed experiments, the plexDIA channel definitions and their isotopologue labels. Because these settings are declared per sample, independent datasets acquired with different scan-range schemes are each parametrised from their own metadata and can be processed together in one invocation, with no change to the workflow. We added a `convert-diann` command to `sdrf-pipelines` that reads these extended columns together with the standard fields (enzyme, fixed and variable modifications, instrument, precursor mass tolerance, fragment mass tolerance, and missed-cleavage settings) and writes a per-run DIA-NN command-line configuration; where SDRF columns are absent, DIA-NN defaults appropriate for the inferred instrument are applied.

2.3. Raw file handling and DIA-NN execution

A key advance over its predecessor `quantms` is that native vendor (raw) formats are first-class inputs in `quantmsdiann`: whereas `quantms` converted every raw file to mzML up front, `quantmsdiann` reads each instrument’s native format directly and converts only when a particular format or DIA-NN version requires it. Thermo `.raw` or `timsTOF` (`.d` or `.d.zip`) files are read natively by DIA-NN $\geq 2.1.0$; for DIA-NN versions lower than 2.1.0 conversion to mzML is performed with `ThermoRawFileParser` [16] (`-mzml_convert`); Bruker `.d` folders are ingested natively, with compressed forms (`.d.tar/.zip/.gz`) decompressed first; and Sciex `.wiff` / `.wiff2` are converted to mzML with the open-source `wiffconverter` (<https://github.com/bigbio/wiffconverter>). Generic `.dia` and pre-converted `.mzML` inputs pass through unchanged, with optional OpenMS re-indexing (`-reindex_mzml`).

2.4. Quality control and QPX export

Per-run and per-cohort quality control reports are generated by `pmultiqc` [19], a proteomics extension of MultiQC [20] that summarises DIA-NN-specific metrics including peptide and protein counts, missed cleavages, charge-state distributions, precursor and fragment mass accuracy, and per-run identification yield. The QPX archive is produced by the `qpxc` CLI from the `qpx` package (<https://github.com/bigbio/qpx>), which writes Parquet files for identifications (peptide-spectrum matches (PSMs), peptides, protein groups, features), metadata (sample, run, ontology), and provenance, and optionally exports MuData (`.h5mu`) containers for scverse-based [21] downstream analysis. The aim of exporting to QPX is to bundle the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival, and to provide a standardised input for downstream tools that support the format (e.g. MSstats, Perseus, or custom scripts).

2.5. Datasets benchmarked and compute environment

Supplementary Note 2 (Table S1) lists all benchmark and reanalysis datasets with their instrument, acquisition mode, and deposition year, and an SDRF prepared from the original sample metadata. They fall into four experiment classes: *ProteoBench* DIA benchmarks (modules 5, 7, 9, and 10); *single-cell and spatial proteomics*, comprising the HeLa single-cell cohort PXD046357 (Orbitrap Astral), the oocyte plexDIA cohort MSV000093870, the PXD064049 spatial Deep Visual Proteomics cohort, and the PXD071075 single-cell scalability sweep; *bulk cell-line panels* (PXD003539, PXD030304, PXD004701, PXD041421, and PXD017199); and *phosphoproteomics* (PXD034128, PXD034623, and PXD049692). For each cell-line reanalysis cohort, the original published quantification was downloaded from a public repository as the comparator. `quantmsdiann` counts follow a single global rule (protein groups at `Lib.PG.Q.Value` ≤ 0.01 , precursors at `Lib.Q.Value` ≤ 0.01 , decoys dropped, no contaminant or positive-quantity filter); the like-for-like-threshold rule applied to each original headline count is described in Supplementary Note 3. Runtime benchmarks were run on the EMBL-EBI HPC cluster (LSF scheduler), with per-task CPU and memory dispatched by Nextflow and memory profiled from the Nextflow trace report (`-with-trace`).

2.6. Benchmarking and cross-study comparison

A distributed workflow is only useful if it reproduces a single-machine DIA-NN run, so we benchmarked `quantmsdiann` against the ProteoBench community submissions [14] on the four public DIA modules (Orbitrap Astral, single-cell Astral, `timsTOF` `diaPASEF`, and `SCIEX` `ZenoTOF 7600`). All `quantmsdiann` runs were library-free (the library predicted in-silico from the FASTA), so for a like-for-like comparison we kept only predicted-from-FASTA DIA-NN community submissions

and report the oldest supported release (1.8.1) and the current 2.5.1-enterprise build at DIA-NN’s recommended per-version q-values under a uniform 1% global precursor cut-off. For the public reanalyses, each deposit’s original published quantification served as the comparator, counted under that study’s own threshold.

Cross-engine identification counts carry three caveats that apply to every comparison in the Results: differences in FDR calibration across search-engine generations, the use of each original study’s own counting criterion, and possible differences in protein sequence database and search settings between the deposited analysis and the reanalysis. These are set out where the comparisons are presented (Results, “The value of reanalysing public DIA deposits at scale”).

3. Results

3.1. *quantmsdiann* pipeline architecture

Built on the nf-core framework [13], *quantmsdiann* accepts a single SDRF file as primary input, which encodes per-sample metadata (cell line or tissue origin, factor values, instrument, post-translational modifications) and per-run file references. We extended the SDRF with DIA-relevant columns (MS1 and MS2 scan ranges, and plexDIA channel definitions and labels) and added an adapter to sdrf-pipelines [12] (*convert-diann*) that translates each run’s declared metadata into a per-run DIA-NN command-line configuration. Every run is therefore parametrised from its own metadata with no manual configuration or parameter-harmonisation step, removing a common source of analysis-script divergence between groups: independent datasets with different scan-range schemes, or a multiplexed plexDIA design, are each analysed directly from their declared SDRF settings without any workflow change. The workflow comprises four mandatory and three optional modules organised around the DIA-NN analysis core (Fig. 1), alternating between per-file parallel stages (red; raw-file preparation, preliminary calibration, and individual search) and collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and *pmultiqc* [19] quality-control reporting) operating on the full cohort.

quantmsdiann follows DIA-NN’s distributed multi-pass flow, alternating parallel per-file work with serial cohort-wide steps (Fig. 1). Each raw file is first prepared *in parallel* (vendor-format conversion, decompression, or indexing) where needed; conversion to mzML is mandatory for early DIA-NN versions and for some vendor formats such as SCIEX *.wiff*. A spectral library is then predicted in-silico from the FASTA once (*serial*; skipped when a library is supplied), and every file is calibrated against it in a *parallel* first pass. The first-pass results are merged into a single empirical library (*serial*); each file is searched again against that library *in parallel*; and a final consolidation pass merges all per-file results into the cohort quantification, applying match-between-runs, normalisation, and protein inference (*serial*). MSstats conversion, *pmultiqc* reporting, and QPX export then run once at the end (*serial*). The two per-file search passes carry essentially all of the compute and scale across cluster nodes, driving the throughput reported below; the serial steps set a largely fixed wall-clock floor. Outputs include the standard DIA-NN report tables: *report.tsv* (or *report.parquet*), *report.pr_matrix.tsv*, and *report.pg_matrix.tsv*, alongside MSstats-formatted input and a *pmultiqc* [19] quality-control report. A harmonised QPX archive bundles identifications, metadata, and provenance into a Parquet-based artefact, queried directly with DuckDB and, for the quantification layers, exported to MuData (*.h5mu*) for scverse-compatible [21] downstream analysis.

Two design choices make *quantmsdiann* a reanalysis platform rather than a single-analysis tool. Because every DIA-NN release ships as a container-pinned profile selectable at runtime, the same SDRF can be analysed with any supported version (1.8.1 through 2.5.1, including the enterprise

build) without reinstallation. Because the per-file work is parallelised, a single invocation scales from six-run benchmarks to cohorts of thousands of runs. We next establish that this distributed execution matches a single-machine DIA-NN run, then use it to reanalyse public deposits with current releases.

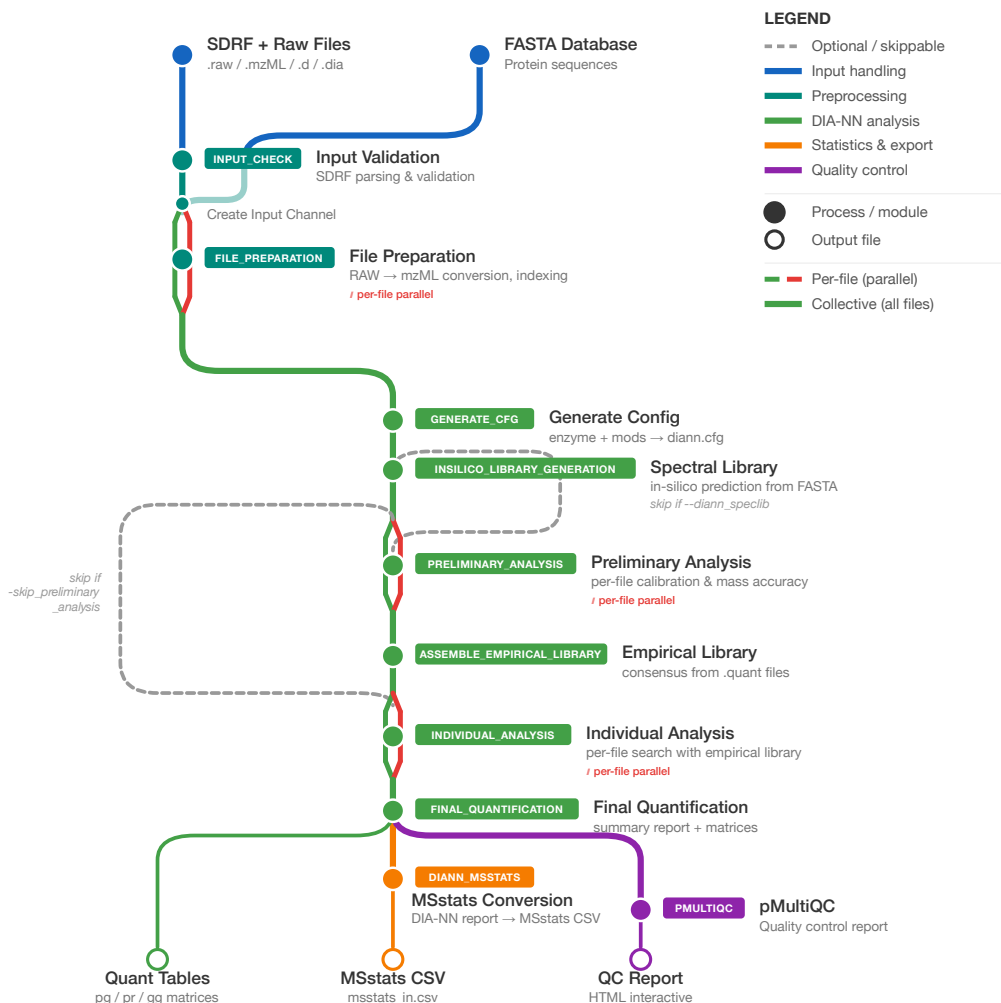


Figure 1: **The quantmsdiann workflow.** Subway diagram: mandatory modules as filled circles, optional/skippable modules as open circles. Per-file parallel stages (red; raw-file preparation, preliminary calibration, individual search) run as one task per raw file, whereas collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, pMultiQC reporting, and QPX export) run once over the full cohort.

3.2. quantmsdiann throughput, scaling, and analytical equivalence

To characterise how **quantmsdiann** scales, we reanalysed an entire single-cell experiment of ~2,300 raw files (PXD071075, Orbitrap Eclipse) across cluster sizes from 10 to 300 nodes and measured end-to-end wall-clock time (Fig. 2a). Because the per-file work is parallelised across nodes, a single SDRF-driven invocation completed the whole cohort within hours, falling from 37.4 h at 10 nodes to 2.2 h at 300. Scaling stays close to the ideal $1/N$ reference up to ~100 nodes and then bends away, with a practical knee near 200 nodes beyond which added nodes return progressively less. This floor reflects the serial cohort-level steps that run once regardless of node count and contention for shared-cluster I/O and scheduler slots, so part of it is specific to this

allocation rather than intrinsic to the workflow. The practical guidance is to size the cluster to the desired turn-around time rather than to maximum throughput.

We then compared time-to-finish across every dataset in the study (Fig. 2b), each reported net of the one-time in-silico library-prediction step, which cannot be parallelised and whose cost depends on the library-generation parameters. (The sole exception is PXD003539, for which the Nextflow execution trace was incomplete and the library step could not be separated out, so its bar conservatively includes it, a ~ 0.4 h overstatement.) Across cohorts spanning three orders of magnitude in file count, wall-clock stayed within hours because the per-file passes absorb additional files in parallel: the 5,798-run ProCan cohort (PXD030304) finished in 6.2 h, in the same band as cohorts orders of magnitude smaller. Time-to-finish is therefore governed by per-run search cost and cohort-level consolidation rather than by file count; the longest reanalysis was a 206-run cohort on an older Q Exactive instrument (PXD017199, 11.4 h), exceeding the far larger ProCan cohort. Peak memory during consolidation stayed within the 64–128 GB available on commodity HPC nodes, so no specialist high-memory hardware is required. The plexDIA and phosphoproteomics cohorts, processed to demonstrate workflow support rather than for cross-engine benchmarking, likewise complete within hours (per-step envelopes in Supplementary Note 4, Figs. S1–S2). To confirm the workflow is not tied to one infrastructure, an independent group reproduced this per-step runtime structure on a separate (non-EBI) SLURM cluster with the same container-pinned profiles (Supplementary Fig. S1b).

With scaling characterised, we asked whether distributed, SDRF-driven execution matches a single-machine DIA-NN run. On the two ProteoBench modules with predicted-from-FASTA DIA-NN community submissions (Orbitrap Astral, $n = 15$; timsTOF diaPASEF, $n = 8$), **quantmsdiann** reproduces the expected HYE per-species $\log_2$ fold-changes, and across both releases its measured ratios sit within the spread of the standalone desktop submissions (Fig. 2c). Quantification accuracy is thus indistinguishable between desktop DIA-NN and **quantmsdiann**.

Identification depth, by contrast, increases with each DIA-NN version on every module: precursors, per-run protein groups, and proteins quantified in every run are all higher for 2.5.1-enterprise than for 1.8.1 (per-run protein-group depth in Fig. 2d; per-version precursor counts and the complete-profile breakdown in Supplementary Note 5, Fig. S3). This tracks the broader ProteoBench community trend [14] and motivates reanalysis of public deposits with the current build.

3.3. Single-cell DIA reanalysis improves with current DIA-NN releases

Single-cell proteomics has rapidly advanced, with significant contributions from computational methods [22] and community recommendations for performing, benchmarking, and reporting experiments [23]. To assess **quantmsdiann** on low-input data, where sensitivity and match-between-runs matter most, we reanalysed two label-free Orbitrap Astral single-cell cohorts, HeLa (PXD046357) and a larger A549/H460 cohort of 146 single cells (PXD049412), through the workflow under DIA-NN 1.8.1 and the current 2.5.1-enterprise build, both parametrised from the same SDRF (Fig. 3). Upgrading from 1.8.1 to 2.5.1-enterprise increased identifications on both cohorts (Fig. 3a). On HeLa Astral, precursors and protein groups each rose by $\sim 17\%$, and the gain held per cell. On the deeper A549/H460 cohort the reanalysis reproduced the study’s reported Astral single-cell depth [4], a median of $\sim 3,400$ protein groups per cell; here the precursor gain exceeded the protein-group gain, consistent with the near-saturated protein depth of this larger cohort. The improvement is largest in the data-completeness profile (the number of protein groups quantified across an increasing fraction of cells; Fig. 3b), consistent with stronger cross-run propagation, while quantitative precision is preserved (Fig. 3c).

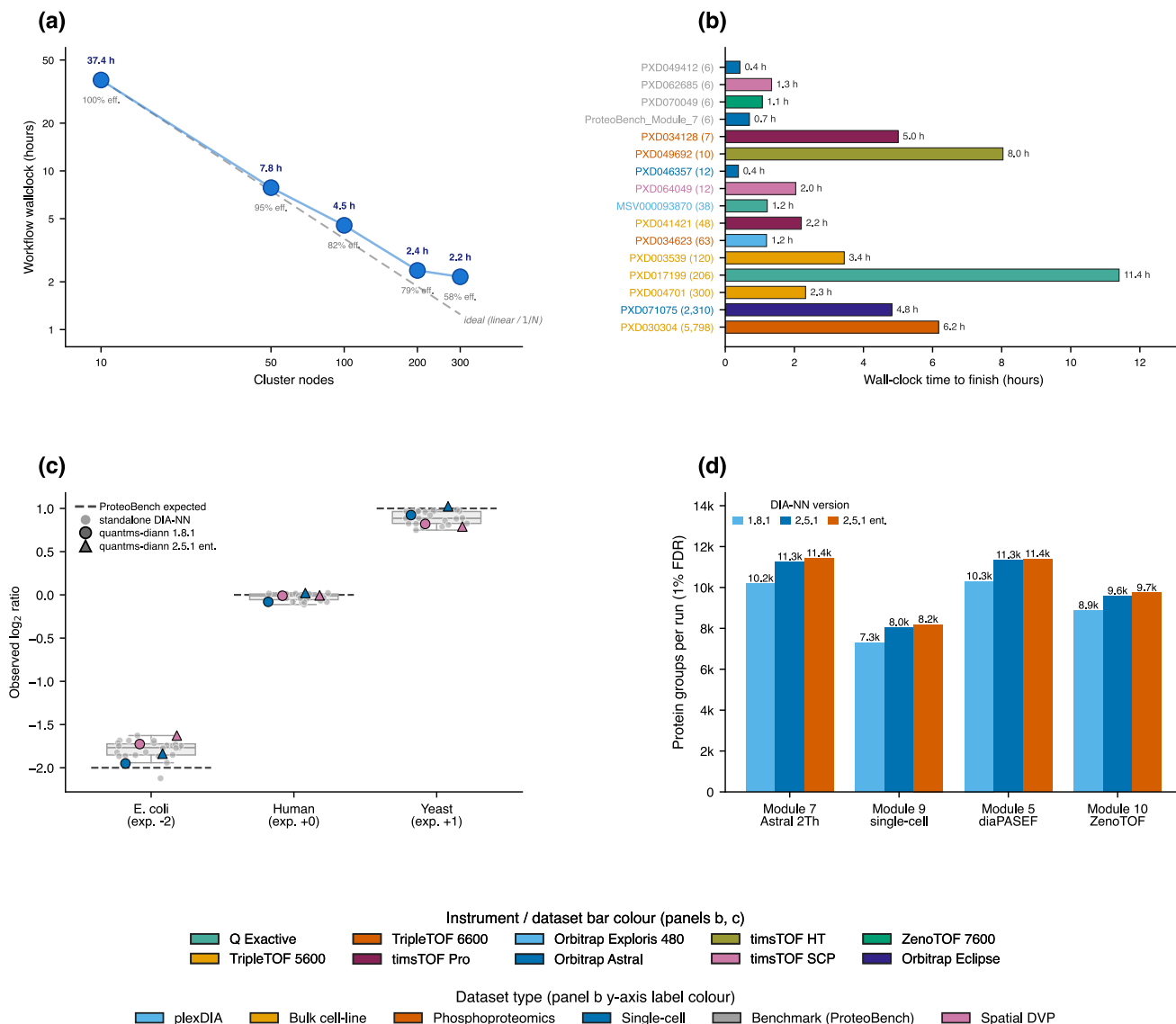


Figure 2: Runtime scaling and ProteoBench quantification accuracy. (a) Workflow wall-clock versus number of cluster nodes (Nextflow `executor.queueSize`) on the PXD071075 single-cell cohort (log-log; one run per point). The dashed grey line is ideal $1/N$ scaling anchored at 10 nodes; parallel efficiency is printed under each dot. The measured curve rises above the ideal line beyond ~ 100 nodes (parallel efficiency falling from 100% to 58%), reflecting fixed serial cohort-level steps and shared-cluster contention on this EMBL-EBI allocation. (b) Wall-clock per reanalysis, one bar per dataset, ordered by cohort size and coloured by instrument family; the dataset-type metadata is shown by the colour of each cohort's y-axis label (descriptive only: bar height reflects cohort size and parallelisation, not sample type per se). (c) ProteoBench accuracy per HYE species: observed log₂ fold-change against the expected ratio (dashed) for the two DIA modules with predicted-from-FASTA DIA-NN submissions (Orbitrap Astral, Module 7; timsTOF diaPASEF, Module 5, PXD062685). Standalone community runs (all versions) form the grey strip and box; `quantmsdiann` points are coloured by instrument and shaped by DIA-NN version (circle, 1.8.1; triangle, 2.5.1-enterprise), tracking the community distribution, i.e. distributed orchestration does not change quantification accuracy. (d) Protein-group identification depth on ProteoBench across three DIA-NN configurations (1.8.1, 2.5.1, and the 2.5.1-enterprise build) and four DIA modalities, as protein groups per run (averaged over the six replicates) at a 1% run-specific protein-group q-value. Depth increases with each release on every module, tracking the accuracy trend in (c); the complete-profile companion (protein groups quantified in every run) and the per-version precursor breakdown are in Supplementary Note 5, Fig. S3.

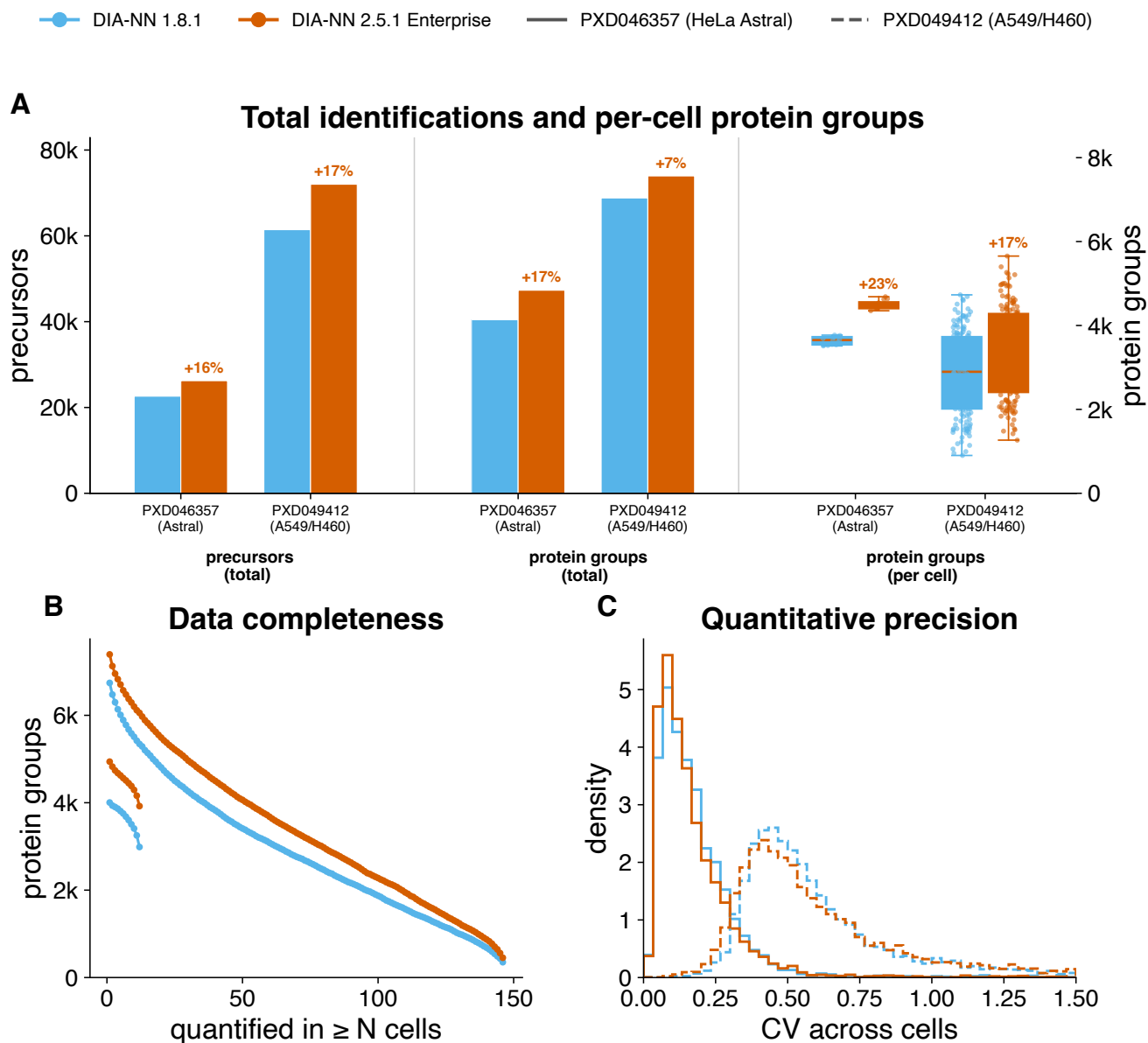


Figure 3: Single-cell DIA reanalysis by quantmsdiann, comparing DIA-NN 1.8.1 with the current 2.5.1-enterprise build on identical raw data. Two label-free Orbitrap Astral cohorts: HeLa (PXD046357) and a larger A549/H460 cohort of 146 single cells (PXD049412). **(a)** Total identifications and per-cell depth per build: total precursors (left axis), total protein groups, and per-cell protein-group distribution (box and jittered points, right axis), with per-build percentage change annotated. A549/H460 reaches comparable per-cell depth on many more cells but is already saturated in total protein groups, so the version effect is concentrated in precursor depth. **(b)** Data-completeness curves (HeLa solid, A549/H460 dashed): protein groups quantified in at least N cells (match-between-runs signature; right endpoint is the complete-profile count). **(c)** Per-protein coefficient of variation across cells (HeLa solid, A549/H460 dashed; axis truncated at $CV = 1.5$), summarising quantitative precision; the larger A549/H460 cohort shows the expected higher cell-to-cell variability.

3.4. The value of reanalysing public DIA deposits at scale

Public DIA deposits are typically processed once, with the search engine available at the time of deposition, and are seldom revisited, leaving recoverable identifications in raw data that has already been collected and published. Reanalysing these deposits with a current DIA-NN build recovers them: across every deposit reanalysed here (single-cell, bulk cell-line, and spatial), reprocessing yielded more protein groups than the originally deposited or published analysis, and more precursors wherever the original was itself DIA-NN (Fig. 4). This gain therefore holds across deposits rather than being specific to any single dataset.

Three caveats bound how these cross-deposit comparisons should be read. First, *FDR calibration*: the comparisons assume that recent DIA-NN (2.5/2.5.1) exercises largely correct false-discovery-rate (FDR) control, whereas older releases (the 1.7-series and 1.8.1) and other search engines may carry substantially higher true FDR at the same nominal 1% q-value [24, 25, 26], and verifying each deposit’s FDR calibration is beyond the scope of this work; the reported gains should therefore be read as identifications recovered under a current, FDR-controlled engine, not as strictly FDR-matched differences. Second, *counting criterion*: for the same reason we compare counts only under each original study’s own threshold, and do not compare plexDIA protein groups or phosphosite numbers across engine versions. Third, *database and search settings*: the reanalysis applies one harmonised FASTA and parameter set across all cohorts, whereas deposited analyses used their own databases and settings, including contaminant databases that are often not deposited (as for the spatial Deep Visual Proteomics cohort, PXD064049); residual count differences attributable to such undocumented choices cannot be fully excluded.

To show the value of running one SDRF-driven workflow across many independent deposits, we reanalysed five public cell-line DIA panels in a single `quantmsdiann` invocation [27, 28, 29, 30, 31]: NCI-60 (originally OpenSWATH), ProCan-DepMapSanger (originally DIA-NN 1.7.x), the Sun PCT-SWATH breast-cancer panel, the MultiPro batch-effect testbed, and Tognetti (67 breast-cancer lines); accessions are listed in the figure legend and Data Availability (Supplementary Fig. S6). For the two cohorts with a published DIA protein-group headline, the reanalysis recovered more protein groups under the original studies’ ≥ 2 -unique-peptide criterion, applied to the deposited count and to the `quantmsdiann` count under its own global protein-group rule: NCI-60 6,812 versus 4,284 (+59%) and ProCan 8,993 versus 6,698 (+34%; Fig. 4a). This ≥ 2 -peptide gain exceeds the change in the total protein-group count: current DIA-NN releases identify more peptides per protein, so proteins that the original analyses supported with a single peptide now clear the stricter two-peptide threshold, expanding the high-confidence protein set rather than only the marginal one. The Sun PCT-SWATH panel, whose deposited OpenSWATH analysis reports proteotypic proteins rather than a ≥ 2 -peptide count, shows a comparable improvement at the standard 1% protein-FDR rule: 8,304 `quantmsdiann` protein groups (`Lib.PG.Q.Value` ≤ 0.01) versus 6,183 in the deposited analysis (+34%). The size of the gain reflects the age and engine of the deposited analysis, largest for the oldest, OpenSWATH-era NCI-60 (+59%); the return is therefore greatest exactly where deposits are most numerous: the large back-catalogue of DIA data processed with older or non-DIA-NN engines. Each original count is recomputed from public data (NCI-60 and Sun from the deposited PRIDE OpenSWATH analyses, ProCan from the authors’ figshare peptide-count matrix), so every bar is reproducible rather than a transcribed paper headline (Experimental Procedures). (PXD041421 and PXD017199 carry no published DIA headline and contribute reanalysis-only counts.) Across the five cohorts the reanalysis spans 12,914 unique UniProt accessions, of which $\sim 44\%$ form a pan-cohort core proteome detected in all five, resolved across 29 tissue categories (Supplementary Fig. S6). Because every count comes from a single SDRF-parameterised invocation, the resulting protein matrix is queryable directly from the published QPX archive and can be filtered or re-grouped by

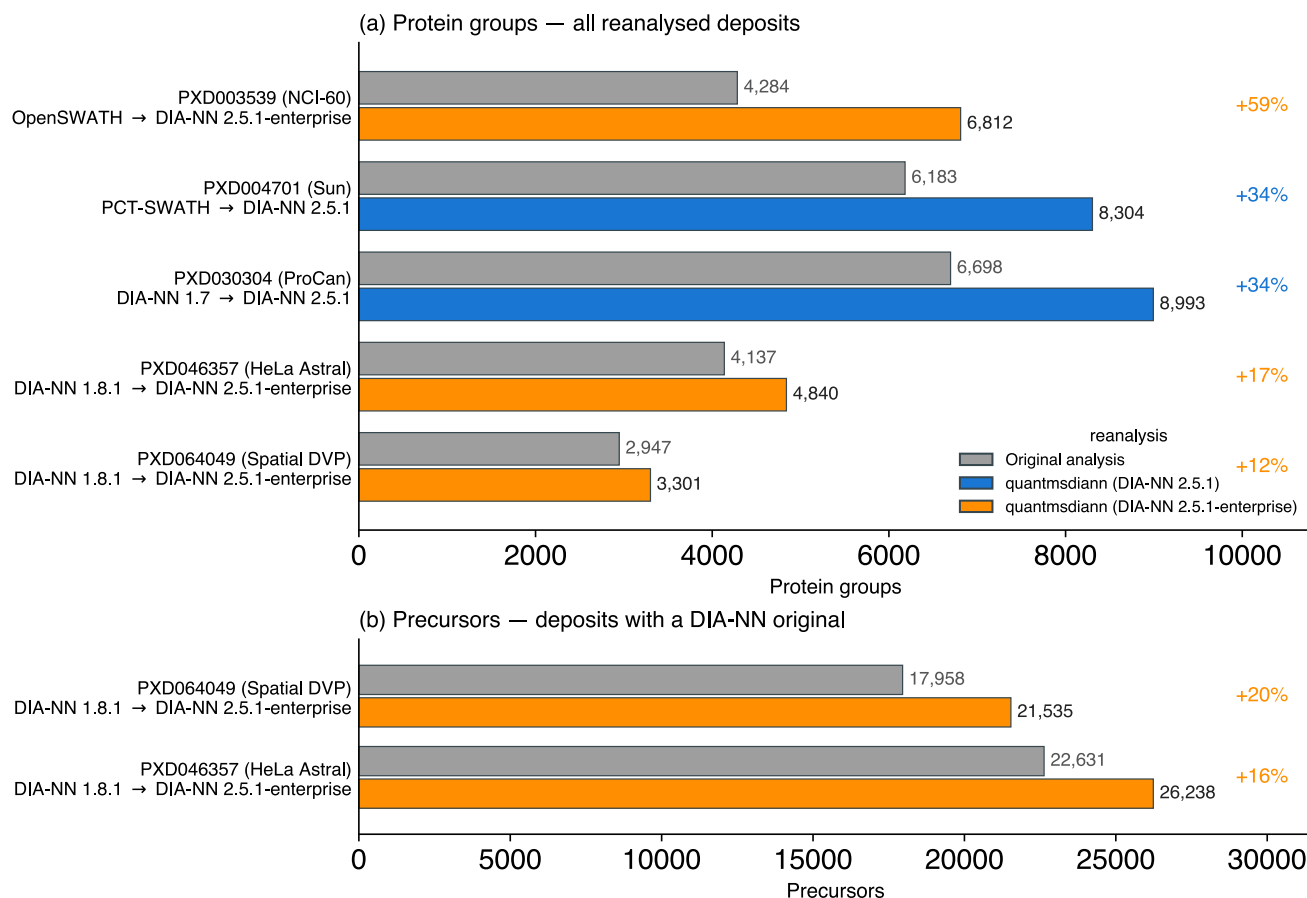


Figure 4: **Reanalysis with current DIA-NN identifies more than the originally deposited analyses.** Each public DIA deposit is shown as the originally deposited or published count (grey) versus the **quantmsdiann** reanalysis (coloured by the DIA-NN version used: 2.5.1 or 2.5.1-enterprise), sorted by gain. **(a)** Protein groups for the five reanalysed deposits, spanning single-cell (HeLa Astral, PXD046357), bulk cell-line (NCI-60, PXD003539; ProCan, PXD030304; Sun, PXD004701) and spatial Deep Visual Proteomics (PXD064049). The multiplexed plexDIA cohort (MSV000093870) is not shown: plexDIA protein-group counts are not comparable across DIA-NN versions. **(b)** Precursors, restricted to the two deposits with a DIA-NN-based original precursor count (HeLa Astral and spatial DVP); the OpenSWATH/PCT-SWATH originals provide no comparable precursor count. The **quantmsdiann** side is counted from the per-precursor report (parquet) for all cohorts; the original side is the deposit's count matrix or published headline, except for the spatial cohort, where both the deposited DIA-NN 1.8.1 and **quantmsdiann** counts are taken from their per-precursor reports under the same global rule (Supplementary Note 3), and over the identical 12 quantified runs on both sides (the deposited DIA-NN log loads additional library/QC runs that are not in the deposited quantification matrix). The five-cohort pan-cohort integration (per-tissue coverage) is in Supplementary Fig. S6.

Cellosaurus, NCIt, or Disease Ontology terms without manual harmonisation, turning independently deposited cohorts into a single reusable cross-study resource. The same pattern extends beyond bulk cell lines to another acquisition mode: a spatial Deep Visual Proteomics cohort of MYCN-amplified neuroblastoma yielded more identifications than its original DIA-NN 1.8.1 analysis (PXD064049; precursors 21,535 versus 17,958, +20%; protein groups 3,301 versus 2,947, +12%, both counted from the per-precursor reports under the same 1% `Lib` q-value rather than the deposited count matrices, which are written at different run-specific q-values per DIA-NN version). Both sides count the identical 12 quantified runs. Notably, the deposited DIA-NN 1.8.1 analysis additionally loaded runs beyond these 12, which would tend to raise its globally identifiable protein count; even with that advantage, the 2.5.1 reanalysis identified more. Residual protein-group inference inflation is also higher in the deposited 1.8.1 analysis (3.2% multi-accession groups) than in the reanalysis (1.3%), so the protein-group gain is a conservative estimate. Per-cohort completeness, peptide-per-protein distributions and accession overlap for the reanalysed cohorts are provided in Supplementary Note 6 (Figs. S4–S6).

4. Discussion

Public proteomics archives such as PRIDE and iProX are increasingly reused for new analyses, rather than accessed only for the studies that produced their datasets. Reuse of repository data can increase statistical power, connect independent cohorts, and test biological signals across laboratories [32, 33]; workflow-based reanalysis has already shown how public deposits can be converted into reusable quantitative resources [9], and resources such as ProteomicsDB show the value of aggregating protein-expression evidence across studies [10]. The need is particularly acute for DIA, a rapidly expanding class of public deposit [8] and a domain in which DIA-NN has become a central analysis engine [1]. Our results show that this archive can be re-mined without changing the analytical meaning of a DIA-NN run: across ProteoBench [14], distributed `quantmsdiann` reproduces the corresponding single-machine result (Fig. 2c), so scaling across a cluster is an infrastructure choice rather than a change in analysis. When applied to public deposits processed with older or non-DIA-NN engines, the same workflow identifies 17% to 59% more protein groups, with the largest gains in older deposits. Successive DIA-NN releases quantify more proteins from the same spectra, so reprocessing legacy data with an improved engine increases the number of quantified proteins, an effect already reported when public datasets were reanalysed with `quantms` [9]. A substantial fraction of the proteome in legacy DIA raw data therefore remains recoverable with current, reproducible analysis.

Identification depth alone, however, does not make archived data reusable. Biological interpretation requires knowing what each sample is, how it was acquired, and how those attributes map onto the quantification matrix; the lack of structured sample metadata remains a major barrier to proteomics reuse [34, 32, 35]. SDRF addresses this problem by making experimental design machine-readable [12]. In `quantmsdiann`, the SDRF drives the analysis rather than annotating it afterwards: every per-run DIA-NN setting is derived from the SDRF, and the same file annotates the resulting QPX archive. This is what allowed the five-cohort cell-line reanalysis to be regrouped by Cellosaurus, NCIt, or Disease Ontology terms directly from the published output, with no manual harmonisation, so separately deposited studies can be queried together.

As DIA experiments grow in sample number, cohort size, and acquisition diversity [23, 22, 6], the bottleneck increasingly shifts from data generation to reproducible analysis. Archive-scale reuse requires thousands of runs to be processed quickly, with quantifications and metadata emitted in standardised, reproducible form. `quantmsdiann` brings together SDRF-driven configuration,

container-pinned DIA-NN profiles, and Nextflow parallelism built on established community infrastructure [11, 18, 13]. In practice, this design reanalysed a 2,300-run single-cell cohort in 2.2 hours on 300 nodes, and assembled a bulk cell-line proteome of 12,914 protein accessions across 29 tissue categories from five independently deposited DIA datasets, each reprocessed in hours (the 5,798-run ProCan panel in 6.2 h); the same workflow also scales down to a small workstation analysis. Because parameters and provenance travel with the SDRF and QPX archive, the output can be reproduced, inspected, or reprocessed with a different DIA-NN version without reconstructing per-study scripts.

By wrapping DIA-NN unmodified and parametrising each run from declared metadata, `quantmsdiann` inherits the engine’s full analytical range rather than being tied to a single acquisition mode or biological application. We demonstrate this across label-free single-cell proteomics, plexDIA, spatial Deep Visual Proteomics, and phosphoproteomics [3, 6, 36, 37]. Other DIA-NN-based applications, including DIA immunopeptidomics [38], follow the same SDRF-driven path. To our knowledge, this is also the first pipeline-level treatment of cross-version DIA-NN reproducibility, a question that will grow as the engine continues to evolve: the same raw data and declared parameters should not lead to different biological conclusions simply because the analysis is rerun later. By design, `quantmsdiann` stops at harmonised quantification and hands batch correction, cell-type assignment, and differential analysis to downstream ecosystems such as `scp` and `scverse-compatible` tools [39, 21], complementing rather than replacing specialised biological analysis workflows.

`quantmsdiann` provides infrastructure for DIA meta-analysis and archive-scale proteomics reuse. It complements earlier community efforts for DIA and quantitative-proteomics workflows [40, 9] by combining current DIA-NN releases, SDRF-first configuration, scalable execution, and QPX export in a dedicated DIA workflow. Applied across the growing public archive, it can integrate independently deposited cohorts into reusable resources and move DIA data towards protein-expression knowledge bases [10] built directly from the community’s accumulated measurements [8]. `quantmsdiann` is open-source under the MIT license and developed within the `nf-core` and `bigbio` proteomics ecosystem, allowing it to track new DIA-NN releases, acquisition modes, and metadata requirements as they appear.

Data Availability

All datasets reanalysed in this study are publicly available through ProteoMeXchange. Cell-line bulk DIA reanalyses: PRIDE PXD003539 (Guo 2019, NCI-60), PXD004701 (Sun 2023), PXD030304 (ProCan-DepMapSanger), PXD017199 (Tognetti 2021), PXD041421 (Wang 2023). Spatial Deep Visual Proteomics reanalysis: PRIDE PXD064049 (MYCN-amplified neuroblastoma, diaPASEF). Scalability experiment: PRIDE PXD071075 (Wu 2025). ProteoBench benchmarks: PXD049412 (single-cell, Module 9), PXD062685 (diaPASEF, Module 5), PXD070049 (ZenoTOF, Module 10), and Module 7 (Astral) datasets. The full list of PRIDE accessions used in the pan-cohort of DIA cell-line reanalyses is provided in Supplementary Note 2 (Table S1). `quantmsdiann` is available at <https://github.com/bigbio/quantmsdiann> under the MIT license; documentation is hosted at <https://quantmsdiann.quantms.org/>. The pipeline is implemented in Nextflow DSL2 ($\geq 25.10.4$) and packages each step as a versioned Docker, Singularity, or Apptainer container image. DIA-NN 1.8.1 is distributed publicly via BioContainers; because the DIA-NN license restricts redistribution of later releases, containers for DIA-NN 1.9 and above are built locally from the recipes in <https://github.com/bigbio/quantms-containers> (see Experimental Procedures). Pipeline version v2.2.0 (release codename “Chongqing”, 2026-06-16) was used for the analyses reported in this paper. Analysis scripts that generated the figures in this manuscript are available

at <https://github.com/ypriverol/quantmsdiann-paper> (this repository).

Supplemental Data

This article contains supplemental data. The Supplementary Notes provide the DIA-NN container-build recipes (Supplementary Note 1), the full PRIDE accession list (Supplementary Note 2, Table S1), the filtering and protein-counting rules (Supplementary Note 3), per-process runtime and resource envelopes (Supplementary Note 4, Figs. S1–S2), the ProteoBench per-version identification breakdown (Supplementary Note 5, Fig. S3), per-cohort completeness, peptide-per-protein and accession-overlap figures (Supplementary Note 6, Figs. S4–S6), and a table of software, formats, and public resources with their repositories (Supplementary Note 7, Table S2).

Conflict of Interest

V.D. holds shares of Aptila Biotech. The other authors declare no competing interests.

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Use of Generative AI

During the preparation of this manuscript, the authors used generative AI tools (Anthropic Claude, and OpenAI Codex and ChatGPT) to assist with software code generation and manuscript editing. All content was reviewed and edited by the authors, who take full responsibility for it.

Author Contributions

QXY, YS, CD and Y.P-R. developed the pipeline. HW helped build the containers and contributed to the workflow implementation and manuscript writing. AL annotated the SDRF metadata and helped run the workflow. TS, FC, NS and JN assisted with the DIA-NN reanalyses and manuscript writing; TS additionally contributed to the workflow design and to the design of QPX and the SDRF format, and JN tested the tool and identified multiple issues. OK provided feedback on the plexDIA datasets. VD provided DIA-NN expertise and guidance on cross-version reproducibility. JAV provided proteomics data-resource infrastructure and funding, supervised the EMBL-EBI contributors, and contributed to manuscript review and editing. MB contributed to manuscript writing and supervised the students; MB and Y.P-R. conceived and supervised the

project. QXY and Y.P-R. wrote the original draft, with input from all authors. All authors read and approved the final manuscript.

Abbreviations

CLI	command-line interface
DDA	data-dependent acquisition
DIA	data-independent acquisition
DIA-NN	data-independent acquisition neural networks (DIA search and quantification software)
diaPASEF	data-independent acquisition with parallel accumulation–serial fragmentation
DVP	Deep Visual Proteomics
EFO	Experimental Factor Ontology
FDR	false discovery rate
HPC	high-performance computing
HUPO-PSI	Human Proteome Organization Proteomics Standards Initiative
HYE	human, yeast, and <i>E. coli</i> (ProteoBench three-proteome mixture)
IQR	interquartile range
LSF	Load Sharing Facility (HPC scheduler)
MS	mass spectrometry
MS1/MS2	first-/second-stage mass spectrometry (precursor/fragment scans)
PASEF	parallel accumulation–serial fragmentation
PSI-MS	HUPO-PSI mass spectrometry controlled vocabulary
PSM	peptide-spectrum match
QPX	Quantitative Proteomics eXchange
SDRF	Sample and Data Relationship Format

References

- [1] V. Demichev, C. B. Messner, S. I. Vernardis, K. S. Lilley, M. Ralser, DIA-NN: neural networks and interference correction enable deep proteome coverage in high throughput, *Nat Methods* 17 (2020) 41–44. doi:10.1038/s41592-019-0638-x.
- [2] F. Meier, A.-D. Brunner, M. Frank, A. Ha, I. Bludau, E. Voytik, et al., diaPASEF: parallel accumulation-serial fragmentation combined with data-independent acquisition, *Nat Methods* 17 (2020) 1229–1236. doi:10.1038/s41592-020-00998-0.
- [3] A.-D. Brunner, M. Thielert, C. Vasilopoulou, C. Ammar, F. Coscia, A. Mund, et al., Ultra-high sensitivity mass spectrometry quantifies single-cell proteome changes upon perturbation, *Mol Syst Biol* 18 (3) (2022) e10798. doi:10.15252/msb.202110798.
- [4] J. A. Bubis, T. N. Arrey, E. Damoc, B. Delanghe, J. Slovakova, T. M. Sommer, H. Kagawa, P. Pichler, N. Rivron, K. Mechtler, M. Matzinger, Challenging the astral mass analyzer to quantify up to 5,300 proteins per single cell at unseen accuracy to uncover cellular heterogeneity, *Nature Methods* 22 (3) (2025) 510–519. doi:10.1038/s41592-024-02559-1.

- [5] J. Wang, Y. Huang, F. Lu, Q. Xu, Z. Yang, Y. Jiang, et al., Benchmarking informatics workflows for data-independent acquisition single-cell proteomics, *Nat Commun* 16 (2025) 10276. doi:10.1038/s41467-025-65174-4.
- [6] J. Derks, A. Leduc, G. Wallmann, R. G. Huffman, M. Willetts, S. Khan, et al., Increasing the throughput of sensitive proteomics by plexdia, *Nat Biotechnol* 41 (1) (2023) 50–59. doi:10.1038/s41587-022-01389-w.
- [7] J. Derks, N. Slavov, Strategies for increasing the depth and throughput of protein analysis by plexdia, *Journal of proteome research* 22 (3) (2023) 697–705. doi:10.1021/acs.jproteome.2c00721.
- [8] Y. Perez-Riverol, C. Bandla, D. J. Kundu, S. Kamatchinathan, J. Bai, S. Hewapathirana, N. S. John, A. Prakash, M. Walzer, S. Wang, J. A. Vizcaíno, et al., The PRIDE database at 20 years: 2025 update, *Nucleic Acids Res* 53 (D1) (2025) D543–D553. doi:10.1093/nar/gkae1011.
- [9] C. Dai, J. Pfeuffer, H. Wang, P. Zheng, L. Käll, T. Sachsenberg, et al., quantms: a cloud-based pipeline for quantitative proteomics enables the reanalysis of public proteomics data, *Nat Methods* 21 (2024) 1603–1607. doi:10.1038/s41592-024-02343-1.
- [10] L. Lautenbacher, P. Samaras, T. K. Schmidt, B. Kuster, M. Wilhelm, et al., ProteomicsDB: toward a FAIR open-source resource for life-science research, *Nucleic Acids Res* 50 (D1) (2022) D1541–D1552. doi:10.1093/nar/gkab1026.
- [11] Y. Perez-Riverol, P. Moreno, Scalable data analysis in proteomics and metabolomics using BioContainers and workflows engines, *Proteomics* 20 (9) (2020) e1900147. doi:10.1002/pmic.201900147.
- [12] C. Dai, A. Füllgrabe, J. Pfeuffer, E. M. Solovyeva, J. Deng, P. Moreno, et al., A proteomics sample metadata representation for multiomics integration and big data analysis, *Nat Commun* 12 (2021) 5854. doi:10.1038/s41467-021-26111-3.
- [13] P. A. Ewels, A. Peltzer, S. Fillinger, H. Patel, J. Alneberg, A. Wilm, et al., The nf-core framework for community-curated bioinformatics pipelines, *Nat Biotechnol* 38 (2020) 276–278. doi:10.1038/s41587-020-0439-x.
- [14] R. Devreese, C. Jachmann, B. Van Puyvelde, H. Webel, W. Bittremieux, C. Chiva, et al., ProteoBench: the community-curated platform for comparing proteomics data analysis workflows, *bioRxivPreprint* (2025). doi:10.64898/2025.12.09.692895.
- [15] UniProt Consortium, UniProt: the universal protein knowledgebase in 2025, *Nucleic Acids Res* 53 (2025) D609–D617. doi:10.1093/nar/gkae1010.
- [16] N. Hulstaert, J. Shofstahl, T. Sachsenberg, M. Walzer, H. Barsnes, L. Martens, et al., ThermoRawFileParser: modular, scalable, and cross-platform RAW file conversion, *J Proteome Res* 19 (2020) 537–542. doi:10.1021/acs.jproteome.9b00328.
- [17] S. Kamatchinathan, S. Hewapathirana, C. Bandla, S. Insua, J. A. Vizcaíno, Y. Perez-Riverol, pridepy: a Python package to download and search data from PRIDE database, *J Open Source Softw* 10 (107) (2025) 7563. doi:10.21105/joss.07563.

- [18] F. da Veiga Leprevost, B. A. Grüning, S. Alves Aflitos, H. L. Röst, J. Uszkoreit, H. Barsnes, M. Vaudel, et al., BioContainers: an open-source and community-driven framework for software standardization, *Bioinformatics* 33 (16) (2017) 2580–2582. doi:10.1093/bioinformatics/btx192.
- [19] Q.-X. Yue, C. Dai, S. Kamatchinathan, C. Bandla, H. Webel, A. Larrea, W. Bittremieux, J. Uszkoreit, T. D. Müller, J. Xiao, J. Cox, F. Yu, P. Ewels, V. Demichev, O. Kohlbacher, T. Sachsenberg, C. Bielow, M. Bai, Y. Perez-Riverol, pmultiqc: an open-source, lightweight, and metadata-oriented QC reporting library for MS proteomics, *Molecular & Cellular Proteomics* 25 (6) (2026) 101530. doi:10.1016/j.mcpro.2026.101530.
- [20] P. Ewels, M. Magnusson, S. Lundin, M. Källér, MultiQC: summarize analysis results for multiple tools and samples in a single report, *Bioinformatics* 32 (2016) 3047–3048. doi:10.1093/bioinformatics/btw354.
- [21] I. Virshup, D. Bredikhin, L. Heumos, G. Palla, G. Sturm, A. Gayoso, et al., The scverse project provides a computational ecosystem for single-cell omics data analysis, *Nat Biotechnol* 41 (2023) 604–606. doi:10.1038/s41587-023-01733-8.
- [22] N. Slavov, Single-cell proteomic technologies: Tools in the quest for principles, *Annual Review of Biophysics* 55 (1) (2026) 253–275. doi:10.1146/annurev-biophys-021424-011450.
- [23] L. Gatto, R. Aebersold, J. Cox, V. Demichev, J. Derks, E. Emmott, A. M. Franks, A. R. Ivanov, R. T. Kelly, L. Khoury, A. Leduc, M. J. MacCoss, P. Nemes, D. H. Perlman, A. A. Petelski, C. M. Rose, E. M. Schoof, J. Van Eyk, C. Vanderaa, J. R. Yates, N. Slavov, Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments, *Nat Methods* 20 (3) (2023) 375–386. doi:10.1038/s41592-023-01785-3. URL <https://doi.org/10.1038/s41592-023-01785-3>
- [24] B. Wen, J. Freestone, M. Riffle, M. J. MacCoss, W. S. Noble, U. Keich, Assessment of false discovery rate control in tandem mass spectrometry analysis using entrapment, *Nat Methods* 22 (7) (2025) 1454–1463. doi:10.1038/s41592-025-02719-x.
- [25] K. K. Krull, S. A. Ali, J. Krijgsveld, Enhanced feature matching in single-cell proteomics characterizes IFN- γ response and co-existence of cell states, *Nat Commun* 15 (1) (2024) 8262. doi:10.1038/s41467-024-52605-x.
- [26] K. Fröhlich, E. Brombacher, M. Fahrner, D. Vogele, L. Kook, N. Pinter, P. Bronsert, S. Timme-Bronsert, A. Schmidt, K. Bärenfaller, C. Kreutz, O. Schilling, Benchmarking of analysis strategies for data-independent acquisition proteomics using a large-scale dataset comprising inter-patient heterogeneity, *Nature Communications* 13 (1) (2022) 2622. doi:10.1038/s41467-022-30094-0.
- [27] T. Guo, A. Luna, V. N. Rajapakse, C. C. Koh, Z. Wu, W. Liu, Y. Sun, H. Gao, M. P. Menden, C. Xu, et al., Quantitative proteome landscape of the NCI-60 cancer cell lines, *iScience* 21 (2019) 664–680. doi:10.1016/j.isci.2019.10.059.
- [28] E. Gonçalves, R. C. Poulos, Z. Cai, S. Barthorpe, S. S. Manda, N. Lucas, A. Beck, et al., Pan-cancer proteomic map of 949 human cell lines, *Cancer Cell* 40 (2022) 835–849.e8. doi:10.1016/j.ccell.2022.06.010.

- [29] R. Sun, W. Ge, Y. Zhu, A. Sayad, A. Luna, M. Lyu, S. Liang, et al., Proteomic dynamics of breast cancer cell lines identifies potential therapeutic protein targets, *Mol Cell Proteomics* 22 (2023) 100602. doi:10.1016/j.mcpro.2023.100602.
- [30] H. Wang, K. P. Lim, W. Kong, H. Gao, B. J. H. Wong, S. X. Phua, T. Guo, W. W. B. Goh, MultiPro: DDA-PASEF and diaPASEF acquired cell line proteomic datasets with deliberate batch effects, *Sci Data* 10 (2023) 858. doi:10.1038/s41597-023-02779-8.
- [31] M. Tognetti, A. Gabor, M. Yang, V. Cappelletti, J. Windhager, O. M. Rueda, et al., Deciphering the signaling network of breast cancer improves drug sensitivity prediction, *Cell Syst* 12 (5) (2021) 401–418.e12. doi:10.1016/j.cels.2021.04.002.
- [32] Y. Perez-Riverol, Proteomic repository data submission, dissemination, and reuse: key messages, *Expert Rev Proteomics* 19 (7-12) (2022) 297–310. doi:10.1080/14789450.2022.2160324.
- [33] S. Hewapathirana, J. Bai, C. Bandla, S. Kamatchinathan, D. J. Kundu, N. S. John, B. Brown-Harry, N. Madhusoodanan, J. M. Riera Duocastella, J. A. Vizcaíno, Y. Perez-Riverol, Quantifying data reuse in proteomics using PRIDE download statistics and a semi-supervised LLM-based framework, *bioRxivPreprint* (2026). doi:10.64898/2026.04.16.718670.
- [34] Y. Perez-Riverol, European Bioinformatics Community for Mass Spectrometry, Toward a sample metadata standard in public proteomics repositories, *J Proteome Res* 19 (10) (2020) 3906–3909. doi:10.1021/acs.jproteome.0c00376.
- [35] E. W. Deutsch, N. Bandeira, Y. Perez-Riverol, V. Sharma, J. J. Carver, L. Mendoza, D. J. Kundu, C. Bandla, S. Kamatchinathan, S. Hewapathirana, et al., The ProteomeXchange consortium in 2026: making proteomics data FAIR, *Nucleic Acids Res* 54 (D1) (2026) D459–D469. doi:10.1093/nar/gkaf1146.
- [36] A. Mund, F. Coscia, A. Kriston, R. Hollandi, F. Kovács, A.-D. Brunner, et al., Deep visual proteomics defines single-cell identity and heterogeneity, *Nat Biotechnol* 40 (8) (2022) 1231–1240. doi:10.1038/s41587-022-01302-5.
- [37] A. Makhmut, D. Qin, S. Fritzsche, J. Nimo, J. König, F. Coscia, A framework for ultra-low-input spatial tissue proteomics, *Cell Syst* 14 (2023) 1002–1014.e5. doi:10.1016/j.cels.2023.10.003.
- [38] J. Scheid, S. Lemke, N. Hoenisch-Gravel, A. Dengler, T. Sachsenberg, A. Declercq, et al., MHCquant2 refines immunopeptidomics tumor antigen discovery, *Genome Biol* 26 (2025) 290. doi:10.1186/s13059-025-03763-8.
- [39] C. Vanderaa, L. Gatto, The current state of single-cell proteomics data analysis, *Curr Protoc* 3 (2023) e658. doi:10.1002/cpz1.658.
- [40] L. Bichmann, S. Gupta, G. Rosenberger, L. Kuchenbecker, T. Sachsenberg, P. Ewels, et al., DIAproteomics: a multifunctional data analysis pipeline for data-independent acquisition proteomics and peptidomics, *J Proteome Res* 20 (2021) 3758–3766. doi:10.1021/acs.jproteome.1c00123.